Full Stack Assignment – 09

Assignment ID: FSD-09
Title: Shift Assignment Management System
Duration: 3 Hours

1️⃣ Objective

Design and develop a **Shift Assignment Management System** that enables staff members to be assigned work shifts while managers control scheduling.
The system must prevent overlapping assignments and ensure **data persistence using a database**.

2️⃣ User Roles

STAFF

- Register and log in to the system
- View assigned shifts

MANAGER

- Log in to the system
- Create and manage shifts
- Assign shifts to staff members
- View all shift assignments

3️⃣ Functional Requirements

Authentication & Authorization

- Implement user registration and login
- Use **JWT-based authentication**
- Protect all backend APIs using JWT
- Enforce **role-based access control** on APIs and frontend views

Shift Management

- MANAGER can create shifts with start and end times
- Shift data must be stored in the database
- Shift assignments must be persisted and retrievable

Shift Assignment

- MANAGER can assign shifts to staff members
- STAFF can view assigned shifts
- Assignment data must reflect real-time database state

4️⃣ Business Rule (Critical)

⚠️ **A staff member must not be assigned overlapping shifts.**

Any assignment that overlaps an existing shift for the same staff member must be rejected.

5️⃣ Technical Constraints (Mandatory)

- All shift and assignment data **must persist in a database**
- Hardcoded or in-memory storage is **not allowed**
- Backend APIs must reflect actual database state
- JWT authentication and role validation are mandatory
- Sensitive configuration must use environment variables

6️⃣ Frontend Requirements

- Login and registration pages
- Role-based dashboards for STAFF and MANAGER
- Shift listing for STAFF
- Shift creation and assignment view for MANAGER

UI styling is not evaluated; correctness and logic matter.

7️⃣ Submission Requirements (Strict)

GitHub Repository

- Backend source code
- Frontend source code
- Proper folder structure
- Clear and meaningful commit history

Live Deployment

- Live frontend deployment link
- Live backend deployment link
- Backend must be connected to the database

README.md (Mandatory)

The README file must include:

- Project overview
- Tech stack used
- User roles and permissions
- API endpoints

- Database schema or collections
- Steps to run the project locally
- Live frontend and backend links

8️⃣ Evaluation Note

The focus is on **time-conflict handling, role enforcement, and end-to-end integration**.